You must not read the question paper before the exam starts.

You must not write anything until the invigilator says that you may start.

Pay attention to the instructions given by the invigilators.

Read the instructions below **carefully** before the exam starts, **check** you know how you are required to complete your answer booklets or sheets.

You must ensure that the front of your answer books/sheets have been completed correctly **by the end of your exam finish time**, no further time will be given.

January 2025 **EC148 2024/5 A 800**

Answer Booklet
Graph paper, A4
Approved calculators are permitted

University of Reading

Introductory Economics (EC148)

Three hours

Answer **all four** questions.
For full marks you need to justify your answers.
This paper carries a total of 100 marks.

Section A
Microeconomics

1. In Wonderland the White Rabbit is using all of his money for pocket watches and umbrellas. For his service at the Queen of Heart's court he receives an income of 2000. Whenever not being asleep Dormouse is the only merchant for pocket watches and umbrellas in Wonderland. He sells pocket watches at 40 per piece and an umbrella costs 8.

   (a) Give the White Rabbit’s budget line.

   (b) If the White Rabbit spends all his income on pocket watches, how many could he buy?

   (c) If the White Rabbit spends all his income on umbrellas, how many could he buy?

   (d) Draw the White Rabbit’s budget set.

   (e) Due to a raw material crisis in Wonderland it becomes more costly to produce umbrellas. As a consequence the unit price of umbrellas in Dormouse’s store increases to 10. Illustrate graphically how this price change affects the White Rabbit’s budget set.

(25 marks)

2. Alice and March Hare are attending the Mad Hatter's tea party. The waiter refuses to offer different drinks per participant so they have to reach an agreement on what to request. The possibilities are: *coffe*e, *tea*, or *wine*. The Mad Hatter proposes the following procedure: they take each pair of alternative drinks and let a majority vote determine their collective ordering. The drink that comes up top shall then be requested from the waiter. The individual rankings (in descending order of desirability, respectively) are as follows:

   - Alice: coffee, tea, wine
   - March Hare: wine, coffee, tea
   - Mad Hatter: tea, wine, coffee

   (a) March Hare suggests that they first consider *coffee* versus *tea*, and then the winner of that contest versus *wine*. Which drink will be ordered?

   (b) Instead, the Mad Hatter suggests that they consider *coffee* versus *wine*, and then the winner of that contest versus *tea*. Which drink will be ordered?

   (c) What order should Alice suggest, if she wants to get her favourite drink served?

   The tea party at the Mad Hatter's house was so wild that the house has become a mess. Since the Mad Hatter very much dislikes tidying up, he now contemplates leaving his house in Tulgey Woods. He finds beautiful mansions for sale in the following areas of Wonderland: Boro Grove, Mallow Marsh, and Whispering Woods. In terms of area the Madhatter prefers Mallow Marsh to Boro Grove, Boro Grove to Whispering Woods, and Whispering Woods to Mallow Marsh.

   (d) Are the Mad Hatter’s preferences about areas complete?

   (e) Are the Mad Hatter’s preferences about areas transitive?

(25 marks)

Section B

Macroeconomics

3. Labour Market and Macroeconomic Variables.

(a) Briefly explain using words, graphs, and equations (if the latter exist) the following concepts.

(i) Real Interest Rate

(ii) GDP Deflator

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) In the real world, a government would like to completely eliminate structural and frictional unemployment.

(b) In an economy, a representative firm, **ProdTech Inc.**, produces goods according to the following Cobb-Douglas production function: $F(K, L) = 3K^{1/3}L^{2/3}$. The firm operates in a perfectly competitive market, meaning it is a price-taker and sells its output at a fixed price of 2.

(i) Derive **ProdTech Inc.**'s profit function and set up the profit-maximization question for the firm to decide how much labour to hire.

(ii) Characterize **ProdTech Inc.'s** labour demand function and graphically illustrate it, if the quantity of capital the firm rents is 1 unit.

(iii) If in this economy, labor supply is given by $L^S = 4W$. Find the labor market equilibrium and illustrate it using a graph. What is the real wage paid to labor? What is the level of unemployment in this economy?

(iv) Suppose the regulator sets a minimum wage of $W = 4$. What happens to labor supply and labor demand as a result of the minimum wage? What happens to the level of unemployment in the economy? Illustrate these changes graphically.

(25 marks)

4. Keynesian IS-LM Model

(a) Briefly explain using words, graphs, and equations (if the latter exist) the following concepts.

(i) Taylor Rule

(ii) Marginal Propensity to Consume

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) A change in the autonomous investment will result in a movement along the PAE curve.

(b) Consider a closed economy of **Macroville** with the following characterization.

- Consumption is characterized by $C = 100 + 0.4Y_D$, where Y_D denotes disposable income
- The investment function is given by $I = 500 + 0.2Y - 100r$, where r is the real rate of interest
- The government only levies taxes on income by the tax rate $\tau = 0.5$ and currently runs a balanced budget
- The demand for money in this economy is given by $(M/P)^D = 0.3Y_D - 100r$
- The nominal money supply is $1{,}200$, and the price level is 3.

Please answer the following questions.

(i) Solve for the IS curve and the LM curve of **Macroville**. Find the slopes of the two curves and illustrate in a graph.

(ii) Based on your answers from (i), what are equilibrium values for all variables in **Macroville**?

(iii) If the **Macroville** government, instead of running a balanced budget, now runs a budget surplus of 100, what would happen to equilibrium output and interest rates (you need only tell me the direction of change and not the exact amount)? Explain your answer.

(25 marks)

(End of Question Paper)